

PAES: AGRICULTURAL STRUCTURES

PAES 401: 2001

Agricultural Structures – Housing for Swine Production

boar

refers to a **male** breeding swine which is at **least 8 months old**

creep area

place for piglets inside the farrowing pen

culling rate

rate of removing undesirable or unproductive animals within the herd

dry sows

unbred sows which have just been weaned; non-pregnant sows

farrowing

act of **giving birth** in swine

farrowing pen

area in which a sow is confined during farrowing and lactation periods, but in which the sow **can turn around**

farrowing stall

farrowing crate device in which a sow is confined during farrowing and lactation periods and which **prevents sow from turning around**

fattener

swine raised for meat production usually starts at **15 kg**

gilt

female swine that has **not farrowed**

grower

swine from **40 to 65 kg**

litter

piglets born in one farrowing

litter index

farrowing index average number of farrowing of one sow per year

occupancy

number of days an animal stays in a pen

sow

any breeding female pig that has farrowed

weaner

weanling piglet that has been recently separated from its mother

LOCATION

building shall be constructed in an **east-west orientation** and the structure for marketable animals shall be located near the service road.

SWINE HOUSING SYSTEM



Figure 1 – One-unit system

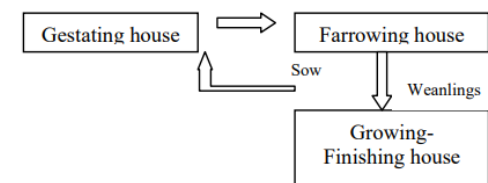


Figure 2 – Two-unit system

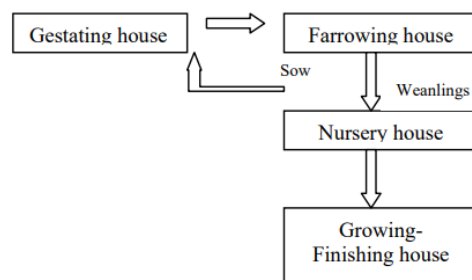


Figure 3 – Three-unit system

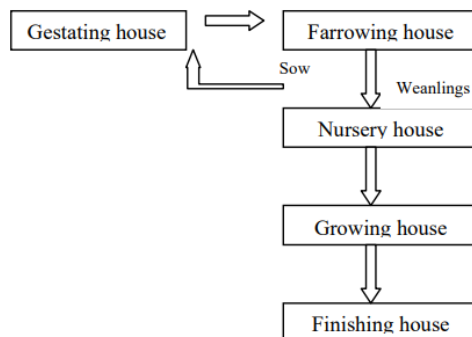


Figure 4 – Four-unit system

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SPACE REQUIREMENT

Table 1 – Minimum recommended space requirement for swine

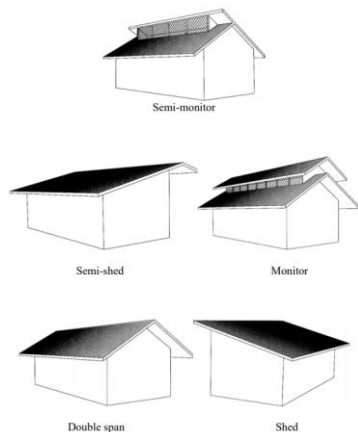
Age and size of animal	Space requirements per animal m ² /animal
Groups of growing swine	
Up to 10 kg	0.11
11 - 20 kg	0.20
21 - 40 kg	0.35
41 - 60 kg	0.50
61 - 80 kg	0.70
81 - 100 kg	0.85
Gilts up to mating	1.00
Adult pigs in groups	2.50

Table 1 (continued)

Age and size of animal	Space requirements per animal m ² /animal
Gestating sows	1.20
Boar pens	7.50
Lactating sows and litters	
Individual pens	7.40
Multi-suckling groups	5.60
Dry sows	1.80

ROOF

At least **one-meter** roof overhang



CEILING

Height shall be at least **2.4 m**

WALLS

Sidewalls have a minimum height of **1 m**

SOLID FLOORS

Skid-resistant with **2%-4%** slope towards gutter, drainage canal or away from the feeding trough.

SLOTTED FLOORS

Table 2 – Recommended slat width and spacing

Size and type of swine	Width of slat mm	Slat spacing mm
Farrowing sows and piglets (up to 30 kg)	18 - 25	8 - 9
Weaners	18 - 25	10 - 14
Finishers	60 - 100	10 - 20
Sows, finishers and boars (over 100 kg)	80 - 125	10 - 25

Partially slotted floors: perpendicular to the long dimension of pens

Totally slotted floor: parallel to the sow

Concrete slats: swine over 30 kg

Expanded metal slotted flooring: under 20kg

PENS

Table 4 – Minimum farrowing crate dimension

Measurement	Dimension m
Length	1.80
Width of stall	0.60
Width of creep	0.50
Height of stall	1.00
Height of creep	0.40

Weaners: slatted floor with slot space clearance of **10-14 mm**

Growing: size should hold not more than **25** growers (**1.5 x 4.8m for 10 growers**).

Finishing: size should hold not more than **25** finishers (**1.5 x 4.8m for 10 finisher**).

Gestating: individual- **0.66 x 1.8m**

Boar: close to dry sow

PEN PARTITION

Range of **160 mm- 210 mm**

Table 5 – Minimum height of pen partition

Swine	Height of pen partition m
Under 25 kg	0.70
25 kg – 100 kg	0.90
Sow	1.00
Boar	1.20

PEN GATE

10 mm iron bars/ **30 mm** pipe

600 mm wide; **0.9-1 m** high

Weanling: **750 mm** high

AISLES and WALKWAYS

Primary: **1.5 m** width

Secondary: **1 m** width

HEATERS

Installed in **creep areas** with **27 C - 35 C** until they are **3 days old**.

Heat lamps: **762 mm** above the floor or **152 mm** above the sow

FEEDING TROUGH

300 mm width; **250 mm** depth

Swine weight	Linear length of trough mm/animal
15 – 25	150
25 – 50	200
50 – 75	250
75 – 100	300
100 - 130	350

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LIGHTING

Table 6 - Recommended lighting intensity for swine housing

Area	Lighting intensity* lux (Lumen/m ²)
Breeding, gestation and farrowing	150
Nurseries	100
Growing and finishing	50
Inspection areas	200

*Refer to Annex C.

VENTILATION

Table 7 – Minimum ventilation rates under normal condition

STAGE	Ventilation rate m ³ /min
Farrowing unit (sow and litter)	0.28
Nursery pens	0.08
Growing-finishing pens	0.12
Breeding and gestating pens (gilts, sows and boars)	0.28

TEMPERATURE

Table 8 – Maximum temperature for housed swine in still air

Class	Temperature °C
Sows and boars	30
Piglets newborn	35
3 weeks	30
Weaners	30
Growers and Finishers	30

STORAGE FOR FEEDS AND SUPPLIES

space required should be based on **two weeks (14 days)** supply of feeds.

PAES 402: 2001

Agricultural Structures – Housing for Broiler Production

brooder guard

materials that are placed **around the brooder stoves** to prevent the chicks from straying too far away from the heat supply until they learn the source of heat

brooding

process of **supplying heat** to the chicks after hatching up to the time that their natural heat regulatory mechanisms become fully functional

litter

material used as **bedding** for animals

open-sided housing

long and narrow type of houses wherein at least one-half of the front and the back of the house are open

enclosed housing

house wherein **inside conditions are maintained** as near as possible to the bird's optimum requirements with the use of mechanical ventilation and artificial lighting

LOCATION

East-west orientation

SPACE REQUIREMENT

Table 1 – Floor space requirements for broiler

Stages	Floor space m ² /100 birds
4 weeks and below	6.25
Above 4 weeks old	12.50

ROOF

1 m roof overhang and should be equipped with gutters

CEILING

Height of **2.4 m**

DOOR

900 mm wide and **2 m high** covered with **25 mm** mesh wire

FLOOR

- Slotted floor**
Width of slat: **25-50 mm**
Spacing: **25-38 mm**
Welded wire: 625mm² mesh every 300 mm
Elevated: **900 mm** from the ground
Posts: diameter at least **200 mm**
- Litter-type floor**
50-100 mm deep, 600 m high

HOUSING SIZES

Width: 10-12 m

BROODING AREA

area shall be surrounded with cardboard, metal sheeting or any hard board with a minimum height of **450 m**

Light

First 48 hrs: 35 W/m²; 3.5 W in each 0.37 m²

After 48 hrs: 10 W/m²; 1 W in each 0.40 m²

Table 2 – Recommended brooding temperature

Age of chicks days	Temperature °C
1 – 7	32 - 35
8 – 14	29 – 32
14 – 21	27 – 29
Beyond 21 days	Provide heat only when necessary

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AISLES and WALKWAYS

Aisles and walkways between doors walls shall be at least **1 m** wide.

FEEDERS

Each feeding trough should have a guard or lip about **10 mm - 20 mm**

Table 3 – Feeder length requirement

Stages	Feeder	
	Linear ^a m/100 birds	Round ^b pieces./100 birds
4 weeks and below	4	4
Above 4 weeks old	7.5	5

^a If both sides of a linear feeder is available to the birds, count each side when figuring available space.
^b 305 mm diameter pan.

WATERER

For every **100 chicks**, two fount-type chick waterers shall be provided for the **first 2 weeks**. After two days the founts shall be placed on stands about **25 mm high**. Each waterer shall hold approximately **3.6 liters of water**

Table 4 – Waterer length requirement

Stages	Feeder	
	Linear ^a m/100 birds	Round pieces./100 birds
4 weeks and below	2.5	1
Above 4 weeks old	4	1

^a If both sides of a linear waterer is available to the birds, count each side when figuring available space.

No bird shall have to go over **3 m to drink water**.

LIGHTING

Artificial lighting with an intensity of **200 lux**

PAES 403: 2001

Agricultural Structures – Housing for Layer Production

litter

material used as **bedding** for animals

litter type

type of flooring utilizing the most commonly available materials such as **rice hull** and **rice straw** and **wood shavings** over the cemented floor

slotted type

type of flooring **with openings** to facilitate cleaning of the droppings

slot-litter type

combination of **slotted and litter type** flooring wherein **slats cover 60%** of the total floor area and **40% is covered with litter**.

community nest

large nesting boxes for **3 - 6 layers**

LOCATION

East-west orientation

SPACE REQUIREMENT

Pen Type

Table 1 – Floor space requirements for layers in pen type housing

Stage	Space requirements m ² /100 birds
Growing (7 – 22 weeks)	
Litter floor	14
Slotted floor	6
Slot-litter floor	7
Laying (beyond 22 weeks)	
Litter floor	17
Slotted floor	9
Slot-litter floor	14

Cage Type

minimum space requirement for birds in cages shall be **5m² /100 birds**

ROOFING

1 m overhang

CEILING

2.4 m high

DOORS

900 mm wide; 2 m high; 25 mm mesh wire netting and be made to swing in both direction

FLOOR

- **Slotted floor**
Width of slat: **25-50 mm**
Spacing: **25-38 mm**
Welded wire: 625mm² mesh every 300 mm
Elevated: **900 mm** from the ground
Posts: diameter at least **200 mm**
- **Litter-type floor**
50-100 mm deep, 600 m high

PEN SIZES

Width: 10-12 m

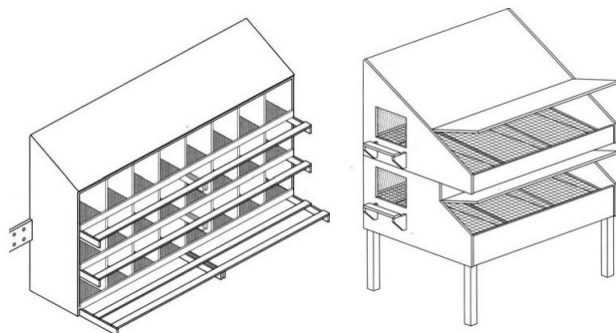
LAYING CAGE

distance spacing between the wires in front of the cage shall be **60 mm** to allow feeding

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floor should have a slope of **21%-25%**. Dimension of the slot should be **15 mm-40 mm x 60 mm**

NEST



a. Individual nest

b. Communal nest

Individual Nests: **310 mm wide, 310 mm high and 310-360 long**. There shall be one nest for each **5-6 layers**.

Community Nest: **600 mm wide, 2.4 mm long, 200 mm opening**. Laying floor shall be sloping **7%** towards the other side. One nest box should be provided for **20-25 layers**.

AISLES and WALKWAYS

clear passage of **800 mm** between cage rows and to the longitudinal walls. At the end walls, **2.4 m** clear passage shall be provided.

FEEDERS

Space requirement: **65 mm/bird**
Each feeding trough should have a guard or lip about **10 mm - 20 mm**

WATERER

Space requirement: **20 mm/bird**
no bird shall have to go over **3 m** to get water

LIGHTING

Artificial lighting with an intensity of **200 lux**

PAES 404: 2001

Agricultural Structures – Housing for Goat and Sheep

buck
mature male goat

doe
mature female goat that has kidded

dry doe
doe **without milk**

kid

young goat under six months old of either sex

ewe

mature female sheep that has **already lambed**

lamb

sheep under six months of age

ram

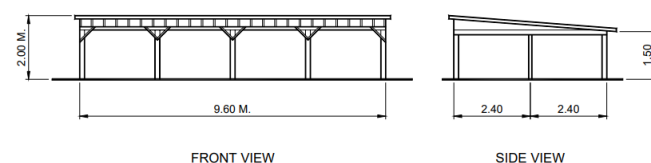
mature male sheep

LOCATION

East-west orientation

TYPE OF HOUSING

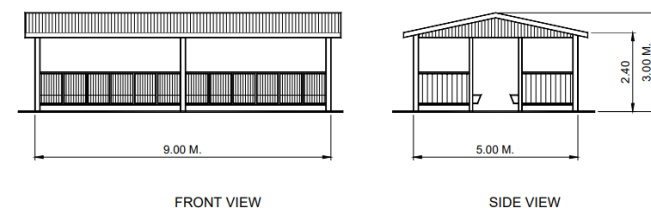
Shed Type



FRONT VIEW

SIDE VIEW

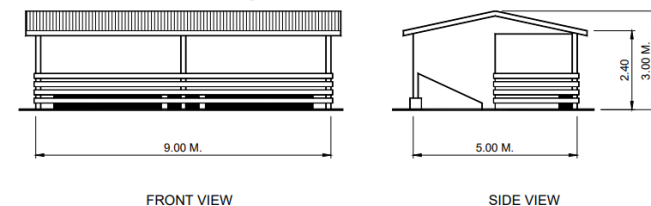
Stall Barns



FRONT VIEW

SIDE VIEW

Confined Housing



FRONT VIEW

SIDE VIEW

SPACE REQUIREMENT

Table 1 – Minimum floor space requirement for goats and sheep in intensive production related to live weight

Animal	Weight kg	Floor space m ² /animal		
		Solid floor	Slotted floor	Open yard
Doe/Ewe	35	0.8	0.7	2
Doe/Ewe	50	1.1	0.9	2.5
Doe/Ewe	70	1.4	1.1	3
Kid/Lamb		0.4 - 0.5	0.3 - 0.4	-
Buck/Ram		3.0	2.5	-

Table 2 – Minimum floor space requirement for pregnant and lactating doe/ewe related to weight

Doe/Ewe	Floor space m ² /animal	
	Pregnant	Lactating
Small (50 kg – 70 kg)	1.3	2.0
Large (over 70 kg)	1.6	2.3

A **fenced loafing** area of **150 m² /50 head** shall be provided adjacent to the animal housing.

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FLOOR

slats should be **70 mm - 100 mm wide, 25 mm - 30 mm thick** and should be laid with **slot space of 10 mm - 25 mm**.

- **staircase type:** minimum step width shall be **800 mm** and the vertical distance between steps shall be **300 mm**
- **slatted floor:** diameter- **5mm**, mesh spacing- **20 mm. 1 m above the ground**
- **concrete floor:** **2% slope; 2-4%**
- **earth floor:** **4%**

ROOF

- **Front eave:** **2m**
- **Rear eave:** **1.5m**
- **Slope:** **not less than 25%**
- **Indigenous materials:** **58% slope**

WALL

Clearance: **150-300 mm**

PENS FOR PEN BARN TYPE HOUSING

- **Height:** not less than **1.2 m**
- **Pen wall & gates:**
 - Metal bars and wooden slats: slot size should be **50-100 mm**
 - Galvanized welded or woven mesh: **100 x 100 mm (post- 70mm dia)**
- **Brooder box:** **500 x 450 x 350 mm**

PEN FACILITIES

- **Depth:** **180-250 mm**
- **Width:** **300 mm**
- **Raised off the ground:** at least **150 mm**
- **Hay racks:** spacing of **130 mm**

Table 3 – Minimum linear feeding space related to live weight

Animal	Weight kg	Feeding space linear mm/animal
Doe/Ewe	35	350
Doe/Ewe	50	400
Doe/Ewe	70	450
Kid/Lamb		250
Buck/Ram		500

WATERING FACILITY

- **open tank drinking system:** **300 mm** of space is required for each **15 - 25 head**
- **automatic watering system:** **1 bowl** or nipple shall be provided for every **50 head**.
- **Free range:** at least **750 mm width**

BREEDING FACILITY

A separate kidding or lambing pen with a minimum dimension of **1 m x 1.5 m** shall be provided.

For every **ten does or ewe**, **one kidding or lambing pen** shall be provided.

LIGHTING

200 lux

FIELD FENCING

- **Height:** **1.5 m**
- **Post** shall be spaced **not more than 4 m** apart and embedded at least **300 mm** into the ground

PAES 405: 2001

Agricultural Structures – Cattle Feedlot

feedlot

area with its physical facilities used for **cattle fattening**

bull

breeding male of any age

calf

young **male or female** under **one year of age**

cow

mature female that has **already calved**

heifer

female between **two to three years** of age which has **not given birth**

SPACE REQUIREMENT

Table 1 – Minimum floor space requirement for cattle

Area	Floor space m ² /animal
Shed area	4
Loafing area	5

FLOOR

- **Concrete floor**
thickness: **76 mm** with **2-4%** slope towards drainage.
- **Earth lot floor:** slope of **4-7%**

ROOF

- **Roof slope:** **25%**
- **Indigenous materials:** **58%**

minimum height of the top of the roof beam shall be **2.5 m** from the floor

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PEN WALL

Diameter of vertical and horizontal railing shall be at least **50 mm and 75 mm** for the post.

The maximum center to center spacing between **vertical railing members** shall be **1.5 m** and for horizontal railing member, spacing shall be **400 mm**

The maximum center to center spacing between **post** shall be **3 m** and shall be embedded in a concrete pedestal with a minimum depth of **400 mm**

Each post shall be provided with **150 mm** concrete protectors.

The pen shall be **1.2 m – 1.5 m** high.

FEEDING TROUGH

400 mm depth, 450 mm – 700 mm bottom width and 700 mm – 900 mm top width

- The **bed of the trough** should be **150 mm** above the level of the apron
- **height from the pen floor** should be **1m - 1.2 m**

The minimum space requirement for **feeding** shall be **750 mm/animal**.

WATERING FACILITY

- **Length of concrete aprons:** **1.5m**
- **Open water tank:** **250 linear mm** for **8-10 heads**
- **Automatic waterer:** **one automatic waterer** should be provided for **15 animals** and it should be placed at a height of **1m**

LIGHTING

200 lux

LOADING RAMP

Loading ramp floor shall have cross battens every **200 mm** to prevent slipping. Ramp should have a **slope of 30%**.

Table 2 – Ramp height for different vehicles

Vehicle	Height m
Gooseneck trailer	0.4
Pick-up truck	0.7
Van-type truck	1.0
Tractor trailer	1.2
Double deck	2.5

- **Permanent:** slope not exceed **36%**
- **Portable or adjustable loading chute:** not exceed **47%**

If stair-stepped concrete ramps are provided, each step should have **90 mm – 100 mm rise** and **300 mm tread width**.

PASSAGE

Central alley or driveway: minimum width of **3 m**

DRAINAGE

e gutter shall have a minimum width of **460 mm** and a minimum depth of **200 mm**

PAES 406: 2001

Agricultural Structures – Cattle Ranch

corral

enclosure for **confining and handling livestock**

holding pen

pen for **confining animals from the pasture**

crowding pen

pen used to **funnel cattle** into the **working chute**

working chute

chute **leading** the cattle from the **crowding pen** to the **holding chute/squeeze**

holding chute

squeeze
use to **restrain animals**

loading chute

chute used to **load cattle** from working chute or crowding pen to a vehicle

LOCATION

Land slope: not more than **58%**

STRUCTURAL REQUIREMENT

- **Corral fence:** **50 x 100-150mm**
- **Four 150 mm** or **five 100 mm** rails should be attached to the inner part of the post with slightly larger spacing at the top of the fence.
- **Space clearance:** **150 mm**
- The minimum **fence height** shall be **1.5 m**.
- For **Brahman** cross and **exotics** breeds, the fence height shall be **1.7 m – 1.8 m**.
- Post should be **150mm - 200 mm** in **diameter** and should be set at least **750**

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mm into the ground. The maximum spacing between posts shall be 2.5 m.

ACCESS ALLEY

shall be at least 3 m wide

HOLDING PEN

at least one holding pen with a size enough to hold the herd

Table 1 – Minimum space requirement for holding pen

Weight of the cattle kg	Space requirement m ² /animal
Up to 270	1.3
270 – 540	1.6
Over 540	1.9

CROWDING PEN

either circular shape (1/4 or 1/2 circle) or funnel shape

- **Funnel shaped:** one straight side and the other side at an angle of 30 degrees. The funnel should be at least 2 m wide.
- **Circular shaped:** have a radius of at least 3 m and equipped with a solid crowding gate constructed out of a 76 mm – 100 mm pipe embedded in concrete.

Table 2 – Minimum space requirement for crowding pen

Weight of the cattle kg	Space requirement m ² /animal
Up to 270	0.5
270 – 540	0.9
Over 540	1.1

- Crowding pen shall have solid walling and the post spacing shall be 1.2 m – 1.8 m. The height of the solid wall shall be 1 m – 1.5 m.

WORKING CHUTE

- **Wall height:** 1-1.5 m high
- **Overall height** including top rail: 1.4-1.8 m post of the working chute should be 1.8 m and should be embedded 0.9 m into the ground.
- Working chute should be curved (C or S-shaped) with maximum curve angle of 15 deg) or offset (offset angle at 30 deg maximum).

Table 3 – Dimensions of working chute

	Dimension m		
	Under 270 kg	270 – 540 kg	Over 540 kg
Working chute with straight side	0.46	0.56	0.71
Working chute with sloping sides			
- Width at 0.8 m height	0.46	0.56	0.71
- Width at bottom	0.38	0.41	0.46

NOTE 1 Add 51 mm to all chute to accommodate bull or large cattle and cows heavy with calf.

HOLDING CHUTE/ SQUEEZE

Either straight or 'V' shaped

Table 4 – Dimension of the holding chute/squeeze

	Dimension m		
	Under 270 kg	270 – 540 kg	Over 540 kg
Height	1.14	1.27	1.27
Width			
- straight sides	0.46	0.56	0.71
- V-shaped sides, width at bottom	0.15 – 0.2	0.2 – 0.3	0.36 – 0.41
Length	1.5	1.5 – 2.4	1.5 – 2.4

LOADING CHUTE

Width shall be 0.66- 0.76 m

Table 5 – Ramp height for different vehicles

Vehicle	Height m
Gooseneck trailer	0.4
Pick-up truck	0.7
Van-type truck	1.0
Tractor trailer	1.2
Double deck	2.5

LOADING RAMP

Cross battens every 0.2 m to prevent slipping with slope of 30%

- **permanently** installed ramp: not exceed 36%
- **portable or adjustable:** not exceed 47%
- **stair-stepped concrete:** 90-100 mm rise and 300mm tread width

FOOTBATH

At least two footbaths of 4.5 long and 0.25-0.3 m deep

DIVISIONAL FENCE

- **Top wire:** 1.4 m high
- **Wire spacing downwards:** 500, 500, 400 mm intervals
- **Fence post:** 1.5 m high; 0.75 m depth
- **Battens clearance:** 75 mm

PERIMETER FENCE

1.5 m high; equally spaced at 0.3 m

PAES 407: 2001

Agricultural Structures – Housing for Dairy Cattle

loose housing

animals are free to move between resting, feeding and watering areas

calf

young male or female under one year of age

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cow

mature female that has already **calved**

yearling

one to two year old cattle of **either sex**

heifer

female between **two to three years** of age which has **not given birth**

parturition

act of **giving birth**

SPACE REQUIREMENT

Table 1 – Minimum floor space requirement for dairy cattle

Class, Age, Size of Animal	Shed or Barn Floor Area m ² /animal
Calves (up to 3 months)	1
Calves (3 - 6 months)	2
Calves (7 months – one year)	3
Yearlings (1 – 2 years)	4
Heifer/Steer (2 – 3 years)	5
Milking and dry cows	6
Cows in maternity stall	10

FLOOR

- **Concrete floor:** thickness- **76 mm**; slope of **2-4%** towards drainage
- **Earth lot:** slope of **4-7%**

ROOF

- **Slope not be less than 25%**
- **Indigenous materials:** **58%** slope

PEN WALL

- **Vertical railing**
 - Diameter: **50 mm**
 - Center to center spacing: **1.5 m**
- **Horizontal railing**
 - Diameter: **75 mm**
 - Center to center spacing: **0.4 m**
- Max center to center **spacing between post:** **3m** embedded in a concrete pedestal with **min 0.4 m depth**
- Each post shall be provided with **0.15 m concrete protectors**
- **Height:** **1.2-1.5 m**

MATERNITY PEN/ CALVING PEN

- Provided for cows that are **two months** away from parturition
- Every **20-25 mature cows**

CALF PEN (up to 3 months)

- Height: **1.2 m**
- Elevated: **0.3 m**
- Slat space clearance: **115-130 mm**
- Space clearance: **25-30 mm**

PEN FOR CALVES (3 months- 1 year)

Up to 6 months

- Horizontal rail: height **0.7 m**

7 months

- Horizontal rail: height **0.9 m**

Yearling, heifer, dry, and milking cow

- Horizontal rail: height **1-1.2 m**

Up to 1 year

- Feed trough: **0.25 m depth, 0.4-0.65 m bottom width, 0.65-0.85 m top width**

Older animals

- Feed trough: **0.4 m depth, 0.45-0.7 m bottom width, 0.7-0.9 m top width**

Table 2 – Minimum feeding space requirement for dairy cattle

Class, age, size of animal	Linear feeding space mm/animal
Calves (3-6 months)	45
Calves (7 months – one year)	50
Yearling, heifer, milking and dry cows Cows in maternity stall	75

WATERING FACILITY

- Slope: **2-4%**

Open Water Tank

- **250 linear mm** for **8-10 heads**

Automatic Waterer

- **One for 15 animals**; height of **1 m**

LIGHTING

- **Feeding area:** **200 lux**
- **Pens:** **70 lux**
- **Feed storage area:** **30 lux**

PASSAGE

- Width: **3 m**

DRAINAGE

- Width: **0.46 m**
- Depth: **0.2 m**

FEED STORAGE

- Horizontal silo: **1:4** wall slope

PAES 408: 2001

Agricultural Structures – Carabao Feedlot

feedlot

area with its physical facilities used for carabao fattening

carabao

Philippine water buffalo or swamp buffalo

PAES: AGRICULTURAL STRUCTURES

LOCATION

East-west orientation

SPACE REQUIREMENT

4 m²/animal

FLOOR

- **Concrete floor:** 76 mm thickness, 2-4% slope
- **Earth lot:** 4-7% slope

ROOF

- Slope: not less than **25%**
- Indigenous materials: min **58%**
- Top roof beam: **2.5 m high** from the floor

PEN WALLING

- Vertical railing: **50 mm**
- Horizontal railing: **75 mm**
- Center to center spacing: **1.5 m**, **0.4** concrete pedestal depth
- Concrete protectors: **0.15 m**
- Height: **1.2-1.5 m**

FEEDING TROUGH

- Dimension: **0.4 m depth**, **0.45-0.7 m bottom width**, **0.7-0.9 m top width**
- Bed of trough: **150 mm** above the level of the apron
- Space requirement: **0.750 m²/animal**
- Horizontal rail: **1-1.2 m** from the floor spaced at 1 m

WATERING FACILITY

- Cylindrical or trapezoidal
- Apron length: **1.5 m**
- Open water tank space: **0.3 m** every **10 heads**

LIGHTING

- Intensity: **200 lux**

LOADING CHUTE

- Width: **0.66-0.76 m**
- Loading ramp: **1.5 m wide** level-loading surface

Table 1 – Ramp height for different vehicles

Vehicle	Height m
Gooseneck trailer	0.4
Pick-up truck	0.7
Van-type truck	1.0
Tractor trailer	1.2
Double deck	2.5

• Loading ramp

- Floor shall have cross battens **every 0.2 m**
- **30% slope**
- **Permanently** installed ramp: **36%**; **portable or adjustable** loading chute: **47%**
- Stair-stepped concrete ramps: **90-100 mm rise**, **300 mm tread width**

PASSAGE

- Width: **3 m**

DRAINAGE

- Width: **0.46 m**
- Depth: **0.2 m**

PAES 409: 2001

Agricultural Structures – Milking Parlor

milking parlor

building or a portion of building **where milking occurs** but where no animals are housed

holding area

area provided to accommodate animals **before milking**

back-out stall

type of stall where animals **must back up to exit the stall**

walk-through

type of stalls that allows the animals to **proceed directly forward after milking is completed**

LOCATION

- **30 m** away from lactating barn
 - Milking frequency (3x a day): **180 m**
 - Milking frequency (2x a day): **275 m**
- **100 m** away from manure piles

FLOOR

- Slope: **2-4%**
- Intersection with the walls: **50-60 mm radius**

WALL

- Intersection with the walls: **50-60 mm**
- Tops and ledges: **40 degrees slope**

PAES: AGRICULTURAL STRUCTURES

CEILING

- **2.4 m** from the finished floor line

WINDOWS

- **16-mesh screen**
- Ledges slope: **45 degrees**
- **1 m** from the finish floor line

DOORS

- Width: **1.5 m**

HOLDING AREA OR WASH PEN

- Space requirement: **2.23 m²/ animal**
- Slope: **2-4%** away from the parlor

ENTRANCE AND EXIT ALLEY

- Width: **0.7-0.9 m**
- Slope: **5%**
- Specific circumstances
 - Slope **greater than 5%** install steps with **150 mm rise** and **400 mm run**
 - Treads may be slope **no more than 1.5%**

MILKING AREA

- **Back-out stall**
 - **0.9 m** headroom
- **Walk-through**
 - **1.8 m** space ahead of the stall
- **Hand milking or bucket milking machine**
 - **1-1.1 m wide**
- **Pipeline-milking system**
 - **0.7-0.8 m**
- Length: **1.5 m**; height: **1.2 m**
- Railings: **50 mm** GI pipe spaced at **0.4 m**
- Milker/ operator and milking equipment spaced at **0.6-0.8 m wide**

Table 1 – Recommended number of milking stalls, machines and operators

Animals	No. of milking stalls	No. of operator		No. of milking machines* (if used)
		Without milking machine	With milking machine	
1-10	1	1	1	1
11-20	2	2	2	1
21-30	3	3	2	1

* Two buckets

- **Milk tank: 2.5 days of milk production**

MILKING MACHINES

- minimum size of vacuum supply pipelines and vacuum pulsator line shall be **50 mm**

LIGHTING

Table 2 – Minimum lighting intensity for milking center

Area	Lighting intensity* lux (Lumen/m ²)
Holding area/wash area	100
Parlor room	
General lighting	200
Milker/Milking operator's area	500

- convenience outlet: installed **1-1.5 m high**

PAES 410: 2001

Agricultural Structures – Lairage for Swine, Small and Large Animals

Lairage

any premise or yard used for the **confinement of animals awaiting to be slaughtered** which include unloading ramp, pens and detention pens

small animals

refers to **sheep, goat, and deer**

large animals

refers to **cattle and carabao**

detention pen

separate compartment in the lairage used to **confine sick or suspected animals**

loose type

animals are **free to move in a pen** while awaiting to be slaughtered

tie-up type

pugnacious animals are **tied within the pen** while awaiting to be slaughtered

slaughterhouse

any building or place used for **killing of animals** where the flesh is intended for human consumption

LOCATION

- constructed at **least 10 m away** from the slaughterhouse
- positioned in the **downwind and downhill** of the slaughterhouse

SPACE REQUIREMENT

- **Large animals**
 - **Loose type: 2.23 m²/ animal**
 - **Tie-up type: 3.30 m²/animal**

PAES: AGRICULTURAL STRUCTURES

- **Small animals**
 - **0.56 m²/animal**
- **Swine**
 - More than 100 kg: **0.7 m²/ animal**
 - Less than 100 kg: **0.6 m²/animal**

UNLOADING RAMPS

- Level surface at the truck tray height (**approx 1.5m**)
- Distance of **2-3m from the truck**
- **Permanent ramp**
 - Catwalks: **610 mm** off the ground
 - Handrail: **1.07 m high**
- **Steps are used**
 - **For cattle**
Distance between steps: **600 mm**
Rise: **150 mm**
 - **For swine**
Distance between steps: **300 mm**
Rise: **90 mm**
- Slope: not the greater than **25 degrees** from the horizontal

PENS

- Separate pens for swine and large animals
- **Elongated pen: 60 degrees** with the pathway

WALLING

- **For cattle pens**
 - GI pipes **50 mm diameter**
 - Set **400 mm apart**
 - Steel post: **100-115 mm diameter**
 - Height: **1.5 m**
- **Hog pens**
 - GI pipe **50 mm diameter**
 - Spacing: **150 mm**
 - Height: **1 m**
- **Small animals**
 - Galvanized welded or woven mesh
 - Mesh: **100 x 100 mm**
 - Galvanized steel: **70 mm diameter**
 - Height: **1.2 m**

FLOORING

- **Cattle pens: 3 degrees slope**
- **Hog pens: 2.4 degrees slope**
- **Large animals and swine**
 - Concrete floors: **30 cm** maximum diamond or square pattern with **50 mm V grooves**
- **Small animals**
 - Galvanized welded mesh **5 mm diameter** and **20 mm spacing**

- **Side curbs** of at least **305 mm in radius**

WATERING AND FEEDING FACILITIES

Water trough for **small animals** shall be placed **250 mm** from the floor to prevent fouling. Feeding through shall be provided above the level of the small animal's head.

LIGHTING

- **110 lux** within the lairage
- Suspect pens: **540 lux**

DRAINAGE

- Drainage line should be covered with iron bars (gratings). Spacing between iron bars should be **95 mm**.

FENCES, GATES, and DIVIDERS

- Round pipe posts shall have a diameter larger than **76 mm**.
- Aisles shall be **1.5m wide**

CHUTE

- **Cattle**
 - Gently **curved or straight race**. It may be **single or double lane**
 - Sides should be **concrete**
 - Final drive of the laneway shall be V-shaped with **800 mm wide** at the top and **500 mm wide** at the floor
- **Swine**
 - Either **double or single race**
 - Sides shall be concrete
 - In double lane, inner should be tubular or **"see through"**

PAES 411: 2001

Agricultural Structures – Slaughterhouse for Swine, Small and Large Animals

slaughterhouse

any building or place used for **killing of animals** where the flesh is intended for human consumption, typical situation and floor plan of a slaughterhouse is illustrated in Annex A

stunning pen

compartment which is suitable for **confining only one animal at a time** while it is being stunned and which is so constructed as to confine, without discomfort, to prevent any

PAES: AGRICULTURAL STRUCTURES

substantial movement of the animal forward, backward or sideways

stunning

renders an animal **insensible** before it is killed

pithing

insertion of a rod or coiled wire through the hole in the **skull of cattle** made by the captive bolt to destroy the brain and spinal cord to prevent reflex muscular action and possible injury to operatives

sticking

severance of the major **blood vessels in the neck** or immediately **anterior to the heart** by means of a **knife and "stuck"** shall be construed accordingly.

bleeding

remove as much blood from the carcass as possible before further handling

scalding

lowering of animal into **steam** to prepare skin for **dehairing**

dehairing

removal of the hair of the carcass

gambrelling

suspending the carcass for particular operation

singeing

cleaning the carcass by **burning the hair**

evisceration

process of **removing the internal organs** in the abdominal and thoracic cavities

dressng

preparation of carcass **after evisceration**, ready for storage or sale

splitting

dividing carcass into parts

carcass

all parts including viscera of slaughtered cattle, sheep, goats or swine that may be used for **human consumption**

meat

edible part of the muscle of cattle, sheep, goats or swine

offal

part of **internal organs** of a slaughtered animal

green offal

digestive tract of ruminants such as the stomach, or the intestines which still contain fecal matter

black offal

digestive tract of swine such as the stomach, or the intestines which still contain fecal matter

detained meat

meat requiring **further examination** as declared by a veterinary inspector after veterinary examination

condemned meat

meat which **is unfit for human consumption** as declared by a veterinary inspector after veterinary examination

gut and tripe

black or green offal

CLASSIFICATION

- **"A" slaughterhouse**
 - **Serve local markets in the community**

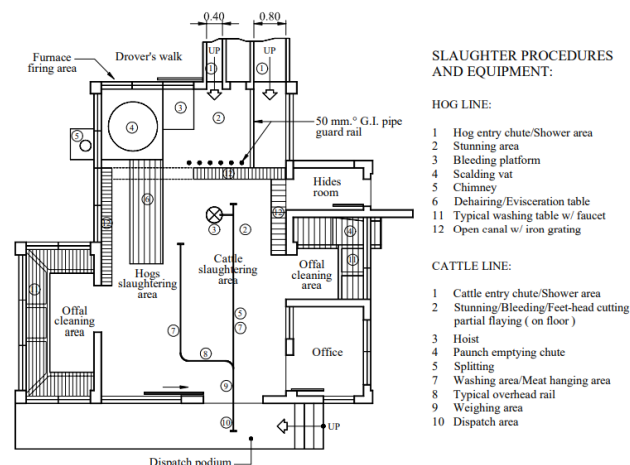


Figure 1 – Typical floor plan and procedures for a class A slaughterhouse

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- “AA” slaughterhouse
 - Serve local markets within the country

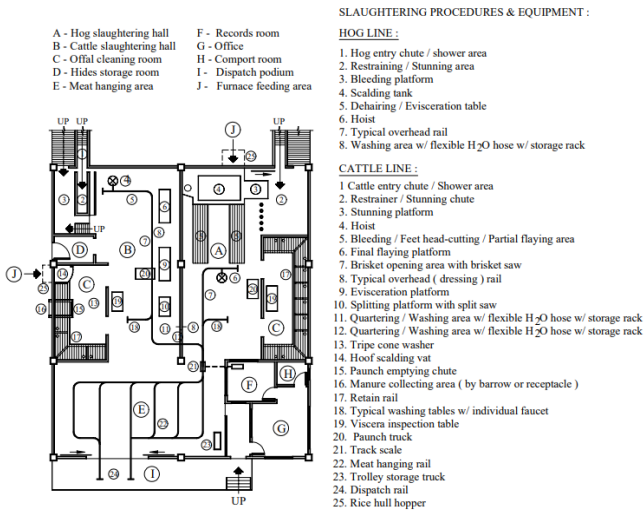


Figure 2 – Typical floor plan and procedures for a class AA slaughterhouse

- “AAA” slaughterhouse
 - Serve any market

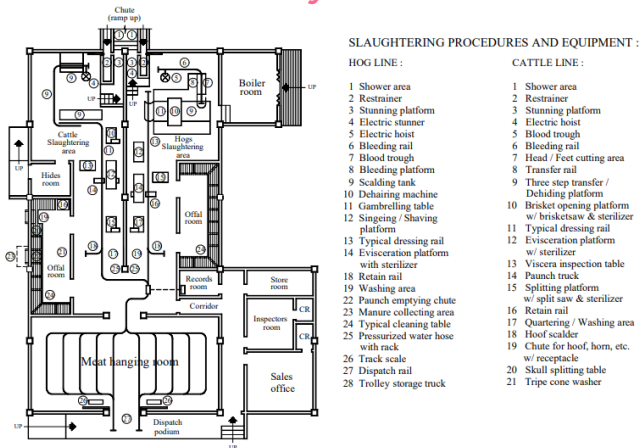


Figure 3 – Typical floor plan and procedures for a class AAA slaughterhouse

LOCATION

- Higher level than its surroundings
- At least **200 m** far from **any buildings**
- Not be located adjacent to the public market areas
- Located near livestock production
- If located near **river, stream, or lake**, slaughterhouse shall be at least **10 m away from the bank**;
- Site shall be dry and well drained
- Area shall be concreted or covered with asphalt
- Sufficient space for possible future expansion

SPACE REQUIREMENT

Table 1 – Recommended dimension for slaughterhouse based on throughput

Animals	Throughput (animals/day)	Dimension (in meters)
Large animals or (small animals)	2	8.6 x 5.8
	(10)	
	30	12.5 x 6.4
	(150)	
Swine	60	23.3 x 15
	(150-350)	
	200	53 x 21.5
	30	8 x 5
	120	12 x 5
	400	19.4 x 8.4

FOUNDATION

- site shall be elevated to at least **600 mm** above the adjacent ground

FLOORS

- Base slab shall be made of concrete (**150 mm thick**) reinforced with **10 mm rods both ways with spacing of 200 mm** or it shall be reinforced with equivalent welded mesh.
- It should be made of strong concrete topping (**40 mm minimum thickness**), or granolithic concrete, or unglazed tiles (ceramic, quarry, cast iron or polymers).
- Slope: **1 degree**

WALLS

- Up to **1.8 m**
- Slope **45 degrees**
- Floor-wall and wall-to-wall junctions with a minimum radius of **50 mm**

CEILING

- At least **3 m** from the floor
- Open joist ceiling center spacing: **900 mm**

WINDOWS

- At least **1.0 m** above the floor
- Slope: **45 degrees**

DOORS and JAMBS

- Doorways: **1.5 m wide**

DRAINAGE

- **Inlet**
 - **40 m²** floor space
 - **300 x 300 mm**
 - Apertures of drain covers: **400 mm²**
- **Drainage line**
 - **20 cm** diameter for ruminal contents

PAES: AGRICULTURAL STRUCTURES

- **15 cm** diameter for stomach contents
- capacity of the **dewatering tank** shall be based on the maximum number of animals to be slaughtered daily at **0.014 m3 per head**

STUNNING PEN

- The area in front of the stunning pen shall be at least **3 m** in width to the opposing wall or bleeding trough and be fitted with upright bars (50 mm in diameter) spaced at **400 mm** in intervals to a height of at least **1.2 m**.
- Illumination: **220 lux**

STICKING and BLEEDING AREA

- **Blood drain: 150 mm diameter**; sloped **170 mm/m**
- **Bleeding trough**
 - Large animals: **1.5 m wide**
 - Small animals and swine: **1.1-1.2 m**

Table 2 – Minimum distance from the carcass suspension contact point to the floor

Animals	Distance (m)
Cattle	3.7
Calves	2.7
Sheep & Goat	2.4
Swine	2.6

BOILER ROOM

- **80 degrees C**

DRESSING AREA

- **540 lux**
- Sanitizing compartment: **82 degrees C**

Table 3 - Minimum distance from the carcass suspension contact point to the floor

Animals	Distance (m)
Cattle	3.1
Calves	2.4
Sheep & goats	2.0
Swine	3.1

GUT ROOM/ TRIPERY

- **220 lux**

CUTTING and BONING AREA

- **220 lux**
- Re-inspection: **540 lux**

STORAGE

- **110 lux**

COLD STORAGE

- **Refrigeration: 2 degrees C**
- **Holding coolers: 4 degrees C**
- **Offal: 1 degrees C or less**

- **Blast freezers: -25 degrees C or lower**
- **Holding freezers: -18 degrees C or lower**
- **Rail spacing**
 - Beef: **900 mm**
 - Pork: **700 mm**
 - Lambs, chevon, venison: **500 mm**
 - Carcass: **300-400 mm**
- **Rail height**

Table 4 – Minimum distance from the carcass suspension contact point to the floor

Animals	Distance (m)
Cattle	3.1
Calves	2.4
Sheep & goats	2.0
Swine	2.4*

*When heads are not removed from dressed carcasses 2.7 m distance is required

DRY STORAGE ROOM

Shelves shall be constructed approximately **250 mm** from the wall and the bottom shall be **600 mm** above the floor.

MANURE AREA

- Exit door: **2.5 m**

PERSONNEL AREA

- **110 lux**

DRESSING ROOM

- **45 degrees slope**
- Floor clearance: less than **350 mm**
- **150 mm** high lockers
- Clothes racks **350-400 mm** floor clearance

INSPECTOR'S OFFICE

- **11 m2** for one inspector and **1.4 m2** for each additional space

NOISE CONTROL

Table 5 – Maximum allowable noise limits (based on background sound levels) for existing sources or places, in dB(A)

Time Period	Dwelling or noise sensitive place	Commercial place
Daytime	55	70
Night time	45	70

*Measurement shall be taken at noise receptors located outside the slaughterhouse boundary.

PAES: AGRICULTURAL STRUCTURES

PAES 412: 2001

Agricultural Structures – Poultry Dressing Plant

carcass

body of **dressed birds**

dressing

process composed of **bleeding, defeathering, eviscerating**, and from which the head, shanks, crop, oil gland and other inedible parts are removed

offal

by-products, organs, glands and tissue other than meat of the food animal

evisceration

process of **removing the internal organs** in the abdominal and thoracic cavities

shackling

process of **restraining birds** prior to slitting

trench drain

trough that **collects the waste** from a larger area and directs the flow to a drain opening

FLOORS

- Slope: **2-4%**
- Intersection with the walls shall be rounded with **50-60 mm radius**

WALLS

- Be painted with white, non-toxic oil or plastic paint
- At least **2 m** of the wall above the floor level shall be tiled or covered with other impervious material.
- All wall tops and ledges shall slope at **45 degrees**
- Coved to the floor-wall and at wall-to-wall junctions with 50-60 mm radius

CEILING

- At least **2.4 m from the floor**

WINDOWS

- Installed with **16-mesh screen**
- Window ledges slope: **45 degrees**
- At least **1 m** from the finish floor line

DOORS

- **External doors** shall be fitted with screens, air flaps or provided with air curtain. The **width** shall be at least **1.8 m**.
- **Internal doors** shall be fitted with self-closing and snug fitting double action doors with a glass (reinforced) panel at eye level or provided with plastic flaps. It shall be at least **1.5 m** wide.

FACILITIES and EQUIPMENT

- Table height: **0.8 m**

LIGHTING

Table 1 – Minimum lighting intensity for dressing plant

Area	Lighting intensity* lux (Lumen/m ²)
Holding area	100
Operating area	
General	200
Working table	500

* Refer to Annex B

- **Convenience outlet: 1-1.5 m high**

WATER SUPPLY AND PLUMBING FACILITIES

- **One hose connection per area**
- One **150 mm - 200 mm drainage** inlet shall be provided for each **37 m2 of floor space**.

PAES 413: 2001

Agricultural Structures – Biogas Plant

biogas plant

plant used to **process animal wastes or manure** to produce biogas and sludge consisting of an inlet/mixing tank, digester, gas chamber and outlet/sludge tank

integrated plant

biogas plant where the **digester and gas chamber form one unit**

split-type plant

digester and gas chamber form separate units

multi-digester plant

plant with **series of digesters**

floating type

plant consisting of digester and a **moving, floating gasholder** that either float directly in the fermenting slurry or in a separate water jacket

PAES: AGRICULTURAL STRUCTURES

fixed type

closed digester with an **immovable**, rigid gas chamber and a displacement pit

balloon type

plant consisting of a **heat-sealed plastic or rubber bag (balloon)**, combining digester and gasholder

collecting tank

holding tank

chamber where **manure and water are collected**, stored and separated from heavy and nonbiodegradable materials before feeding them into the digester

inlet pipe

serves as **conveyor of the manure-water mixture** or slurry **from the mixing tank to the digester**

digester

biodigester

bio-reactor

anaerobic reactor

any water and air tight container designed for the **process of anaerobic microbiological degradation of organic matter** into which the slurry is introduced for digestion and methanization

baffle board

division in the digester that prevent the slurry from premature exit into the sludge/outlet tank

stirrer

mixer

agitator

mechanical device inside the digester used to **stir the slurry**

gas chamber

space inside or outside the digester for the collection and storage of biogas

gasholder retainer

cantilever beam that holds the gasholder/movable cover **in position** at the desired biogas pressure

outlet pipe

serves as conveyor where **the effluent** or the slurry is forced out

backfill

layer of compacted soil and gravel to support the digester wall

loading rate

amount of **slurry fed per unit volume** of digester capacity per day

substrate

organic material used to produce biogas

seeding

adding or introducing anaerobic bacteria to the digester

slurry

mixture of manure and water

freeboard

difference in height **between the digester wall and the filling line**

filling line

level of slurry when the digesters is at full load

retention time

average period that a given quantity of slurry is **retained in the digester** for digestion

toxic materials

materials that inhibit the **normal growth of pathogens** in the digester such as mineral ions, heavy metals and detergents

methanization

digestion

various processes that take place among the **methanogens, non-methanogens and substrates** fed into the digester as inputs

methanogens

anaerobic bacteria that act upon organic materials and in the process, produce biogas

mesophilic temperature rage

temperature range of **20C– 40C** where mesophilic bacteria operates

gas production rate

amount of **biogas produced per day per cubic meter** of slurry

biogas

mixture of gas (composed of 50 to 70 percent methane and 30 to 40 percent carbon dioxide) produced by methanogenic bacteria

PAES: AGRICULTURAL STRUCTURES

scum

layer of **floating material** (mainly fibrous) on the slurry

sludge

settled portion or precipitate of the slurry; a mud-like, semi-solid mass

effluent

residue that **comes out at the outlet** after the substrate is digested/processed inside the digester

LOCATION

- It should be located **as near as possible to the animal pen** and should be lower than elevation of animal pen canal.
- Soil foundation should be stable and away from tree roots intrusion.

COLLECTING TANK VOLUME

- Continuous-fed storage: size of the tank should not exceed to the total slurry volume **for 10 days**
- The slurry volume is the volume occupied by the manure and water at a ratio **of 1:1 (1 kg of manure: 1 L of water)**.
- Batch-fed: slurry input rate shall be multiplied by the interval of slurry charging.

Table 1 - Estimated daily quantity of available manure

Animals	Manure available kg/day/animal
Pigs	
Porker, 3-8 months old, mixed ages	2.20
18-36 kg	2.55
36-55 kg	5.22
55-73 kg	6.67
73-91 kg	8.00
Cow	
Feedlot animal	14.0
Breeding animal	13.0
Work animal	7.50
Buffalo	
Breeding animal	14.0
Work animal	8.00
Horse	
Breeding animal	13.50
Work animal	7.75
Chicken	
Layer, 6 months or older	0.075
Broiler, day-old to 8 weeks	0.025

INLET PIPE

- Minimum diameter: **0.2m**

DIGESTER VOLUME

- digester tank capacity is calculated from the daily slurry volume multiplied by the retention time

Table 2 - Retention time for animal manure for mesophilic temperature range

Substrate	Retention time days
Liquid pig manure	15 - 25
Liquid cow/carabao manure	20 - 30
Liquid chicken manure	20 - 40
Animal manure mixed with plant material	50 - 80

OPTIMUM CROSS-SECTION OF A DIGESTER PLANT

Floating Type

- Baffle Board:** The height should range from **25% - 50%** of the height of the filling line. The height of the filling line should be calculated by subtracting the freeboard from the digester height. If there is no freeboard, filling line is equal to digester height.

Fixed Type

- The digester and gas chamber is integrated into one tank. The total height of the plant depends on the sum of the digester and gas chamber volume. **Eighty percent** of the total digester volume is occupied by the slurry.

Balloon Type

- The digester volume is **80%** of the total digester volume is occupied by the slurry.

Table 3 - Optimum height/length ratios of digesters and tanks (freeboard excluded) for volume up to 70 m³ and wall thickness of up to 25 cm

Horizontal Cross-section	Height/Length Ratio, r (Height/Diameter or Height/side)	
	Floating Type (Integrated) Plants and Open Tanks	Floating (Separate Gasholder) and Fixed Type Plants
Circular	0.500	1.00
Square	0.500	1.00
Rectangular ^a		
L = 1.2W	0.455	0.91
L = 1.4W	0.420	0.84
L = 1.6W	0.385	0.77

^a Coefficient of W is the desired length/width proportion, p

Horizontal Cross-section	Height/Length Ratio, r (Height/Diameter or Height/side)	
	Floating Type (Integrated) Plants and Open Tanks	Floating (Separate Gasholder) and Fixed Type Plants
Rectangular ^a		
L = 1.8W	0.360	0.72
L = 2.0W	0.340	0.68
L = 2.5W	0.295	0.59
L = 3.0W	0.260	0.52
L = 3.5W	0.235	0.47
L = 4.0W	0.215	0.43
L = 5.0W	0.185	0.37

^a Coefficient of W is the desired length/width proportion, p

GAS CHAMBER VOLUME

- The volume of gas production potential should be calculated by multiplying the **amount of manure by its gas production rate**
- Floating type:** It should be calculated by getting the product of accumulation rate of biogas and the longest duration when all non-continuous devices are simultaneously idle. The product should be multiplied by **1.3** to account for the **30%** fluctuation in biogas production.
- Fixed type and balloon type, about **20%** of the total digester volume is occupied by the gas generated.

PAES: AGRICULTURAL STRUCTURES

OUTLET PIPE

- Minimum diameter shall be **0.2 m**

OUTLET TANK VOLUME

- Floating and balloon type: minimum volume shall be equal to the daily slurry input of the digester.
- Fixed type: volume shall be **1/3** of the digester volume occupied by the slurry

COLLECTING TANK

- Concrete channels minimum slope: **2%**
- The floor of the mixing tank should be inclined from **8.5% - 17.5%** toward the inlet pipe and it should be elevated at least **0.2 m** from the filling line.

INLET PIPE

- Concrete pipe (prefabricated RC pipe) should be used and it should be inclined **58%** with digester wall.
- If no retainer, lower end should be located **100 mm** from the floor of the digester.

DIGESTER

Fixed and Floating Type

- Reinforced digester: reinforcement diameter **100 mm** RSB spaced at **0.15 m**
- Plaster for concrete walls in **three layer (13 mm, 6 mm, and 6 mm thick)**
- Manhole cover should be **0.5 m** in diameter and **0.125 m** thick

Balloon Type

- Floor slope: **1.5%**
- Plastic tube: **200 microns**

AGITATOR/STIRRER

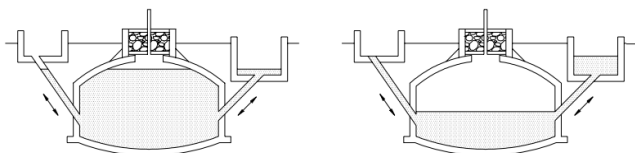


Figure 1 – Mixing substrate through inherent flow in fixed-dome plants

GAS CHAMBER

Floating-Drum Plant

- gas drum should consist of **2.5 mm** mild steel sheets for the sides and **2 mm** sheets for the top.
- Sloping roof of **16.5%**
- depth of liquid jacket should be about **95%** of the height of the gasholder and it shall not be less than **300 mm**.

Fixed Dome Plant

- Plaster shall be applied in three layers (**13 mm, 6 mm, and 6 mm thick**).
- The gas chamber shall be capable of withstanding an internal pressure of **0.15 bar**.

GAS OUTLET PIPE

- The T-joint is inserted in the bottle and water is added to a depth of **40mm - 50 mm** above the lower point of the T
- PVC should be laid at least **0.25 m** deep underground

Table 4 - Pipe diameter for different pipe lengths and flow-rate (maximum pressure loss <5 mbar)

Length m	Galvanized steel pipe mm			PVC mm		
	20	60	100	20	60	100
Flow-rate m ³ /h						
0.1	12.7	12.7	12.7	12.7	12.7	12.7
0.2	12.7	12.7	12.7	12.7	12.7	12.7
0.3	12.7	12.7	12.7	12.7	12.7	12.7
0.4	12.7	12.7	12.7	12.7	12.7	12.7
0.5	12.7	12.7	19.0	12.7	12.7	12.7
1.0	19.0	19.0	19.0	12.7	19.0	19.0
1.5	19.0	19.0	25.4	12.7	19.0	19.0
2.0	19.0	25.4	25.4	19.0	19.0	25.4

OUTLET PIPE

- Concrete pipe (prefabricated RC pipe) should be used with a minimum diameter of **200 mm** and it should be inclined **58%** with digester wall.

OUTLET TANK

- Overflow should be provided with the height of at least **100 mm** lower than the lower surface of the gas chamber.
- The height of the floor of the chamber from the filling line shall be at least equal to the operating pressure for appliances using biogas or the height should be at least **0.2 m** from the filling line plus **15% freeboard**.

GROUND WATER DRAINAGE

- In case groundwater is in great quantity, deep wells should be constructed **2 m** away from the excavated pit, with a depth **0.8 m – 0.1 m** lower than the pit, then pump water away from wells.

PUMPS

- Centrifugal pumps for liquid substrate or rotary pumps for substrate of less than **8%** solid content or positive displacement pumps for substrate with higher solid content may be used

WATER HOLDING TANK

- Digester should be filled with water and let it set for **3-5 hours**

PAES: AGRICULTURAL STRUCTURES

DIGESTION PROCESS

Stage 1 – Hydrolysis

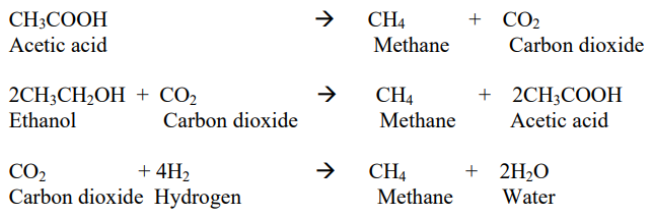
solubilized into simpler ones with the help of extracellular enzyme released by the bacteria.

Stage 2 – Acidification

fermented under anaerobic condition

Stage 3 – Methanization

processed by methanogenic bacteria to produce methane.



PARAMETERS

- **Temperature:** mesophilic temperature **20-40C** with temperature
- **pH: 7-8.5**, if pH value **drops below 6.2**, the medium will have a toxic effect on the methanogenic bacteria
- **Available nutrient**

Table B.1 – Nutrient content of common animal excrements

Animal	P ₂ O ₅		K ₂ O	
	kg/a	%	kg/a	%
Cow	34	0.2	84	0.5
Pig	56	0.4	35	0.3
Chicken (fresh droppings)	194	1.0	108	0.6
Chicken (dry droppings)	193	4.6	106	2.5

- **Inhibitory factors**

Table B.2 - Limiting concentrations for various inhibitors of biomethanization

Substance	Concentration mg/l
Copper	10 – 250
Calcium	8000
Sodium	8000
Magnesium	3000
Nickel	100 – 1000
Zinc	350 – 1000
Chromium	200 – 2000
Sulfide (as sulfur)	200
Cyanide	2

- **Nitrogen Content and C/N ratio:** should be within the range of **1:20 – 1:30**. The C/N ration should not be more than **1:35**

GAS PRODUCTION POTENTIAL

Table B.4 –Gas production potential of various types of manure in m³/kg

Manure	Retention Period days			
	25	30	35	50
Pig	0.058	0.063	0.068	0.077
Cow	0.030	0.034	0.037	0.043
Buffalo	0.030	0.034	0.037	0.043
Horse	0.045	0.051	0.056	0.065
Chicken	0.060	0.065	0.069	0.078

EXCAVATION AND BACKFILLING

- Should be as **vertical** as possible. **0.15 m** minimum gap on both sides should be allowed for backfilling.

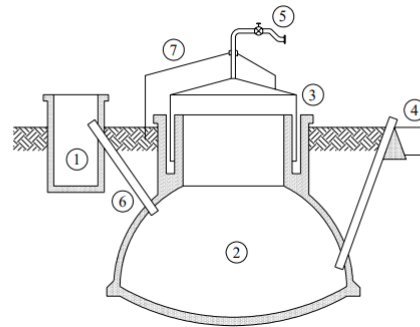
- **Loose soil: 30% gravel, 70% soil, water 20-25%**

Table C.1 - Maximum excavation slope on various grounds

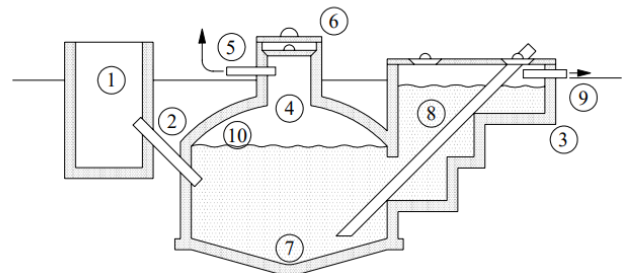
Kind of soil	Ratio of height to width
Sandy soil	1:1
Clayey sandy soil	1:0.67
Clayey soil	1:0.50
Clay	1:0.33
Soil with gravel	1:0.67
Dry loess	1:0.25

PARTS OF BIOGAS PLANT

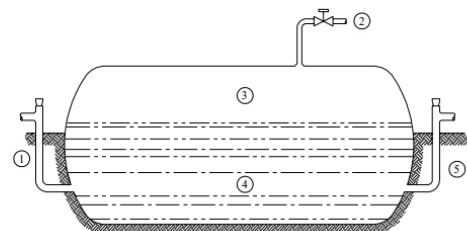
D.1 Parts of Floating type plant with external guide frame: 1. Collecting pit, 2. Digester, 3. Gas chamber, 4. Slurry store, 5. Gas pipe, 6. Fill pipe, 7. Guide frame



D.2 Parts of Fixed type plant: 1. Collecting tank with inlet pipe and sand trap, 2. Digester, 3. Compensation and removal tank, 4. Gas chamber, 5. Gaspipe, 6. Entry hatch, with gas tight seal, 7. Accumulation of thick sludge, 8. Outlet pipe, 9. Reference level, and 10. Supernatant scum, broken up by varying level.



D.3 Parts of Balloon type plant: 1. Inlet, 2. gas pipe, 3. Gas chamber, 4. Slurry volume and 5. Outlet pipe



PAES: AGRICULTURAL STRUCTURES

PAES 414: 2001

Agricultural Structures – Waste Management Structures Part 1: Agricultural Liquid Waste

aerobic

requires **free oxygen**

agricultural liquid waste

consist of **liquid waste and slurry** resulting from the production of livestock and poultry; and processing of crops, livestock and poultry

anaerobic

presence of free **oxygen** is **not** required

clean runoff

runoff **not contaminated with manure** such as runoff from roofs, grassed areas, drives and other areas which are not animal alleys

disinfection

process of **killing pathogenic microorganisms**

dissolved solids

part of total solids **passing through the filter** in a filtration procedure

effluent

liquid waste, partially or completely treated, **flowing out** of a reservoir, basin, or wastewater treatment plant

facultative lagoons

lagoons that can function as **aerobic or anaerobic** depending on the environment

fixed solids

part of total solids remaining **after volatile gases driven off at 600C**

grit

non-biodegradable component of liquid waste composed of sand, gravel, cinders or other heavy solid materials

holding pond

storage where liquid waste is **stored before final disposal**

influent

liquid that **flows into** a containing space

lagoon

pit in the ground where liquid waste is stored to produce a higher quality effluent

liners

system of clay layers and/or geosynthetic membranes used to contain leachate and reduce or prevent contaminant flow to groundwater

lot runoff

rainfall containing animal manure

manure

accumulated **moist animal excrement** that does not undergo decomposition or drying; it include feces and urine which may be mixed with bedding material, spilled feed or soil

pathogenic microorganism

microorganism **capable of causing diseases**

primary treatment

treatment that causes substances in **liquid waste to readily settle or float**

secondary treatment

treatment used to convert dissolved or suspended materials into a form more readily separated from the liquid waste being **treated**

sludge

precipitate resulting from coagulation or sedimentation of liquid waste

slurry

watery mixture of insoluble solid

suspended solids

solids removed **by filtration**

total solids

residue remaining after water is removed from waste material by evaporation

volatile solids

part of total solids **driven off as volatile gases** when heated to 600C

5-day Bio-Chemical Oxygen Demand (BOD5)

quantity of oxygen needed to satisfy **biochemical oxidation** of organic matter in waste sample in **5 days at 20C**

PAES: AGRICULTURAL STRUCTURES

TYPICAL PROCESSING DIAGRAMS FOR LIQUID WASTE

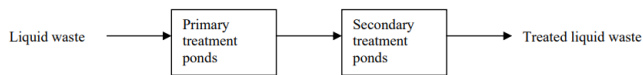


Figure 1 – Natural biological treatment in ponds

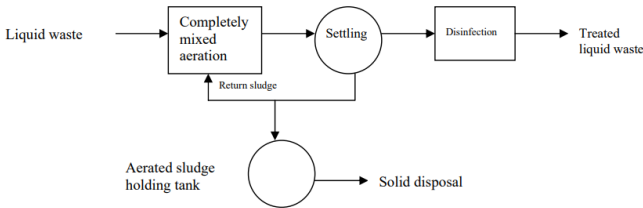


Figure 2 – Biological processing without primary sedimentation

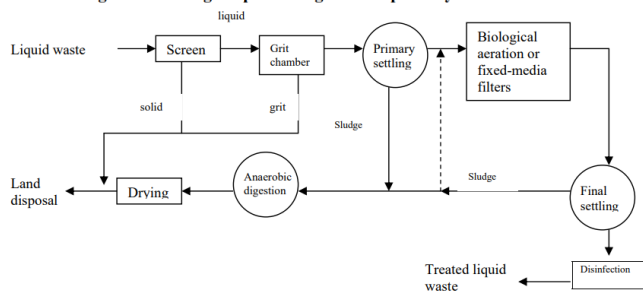


Figure 3 – Conventional liquid waste treatment plant

LOCATION

- Shall conform to **DENR AO 34 and 35, PD 1067, and PD 1152**
- Constructed on soils with at **least 15% clay content**

WASTE COLLECTION

Alleys

- **Scrape Alley**
 - Dairy and Beef Cattle: **2.5- 4m wide**
 - Swine and Poultry: **1- 2.5 m wide**
- **Flush Alley**
 - **1- 3 m**
 - Initial flow
 - Underslat alley: **75 mm**
 - Open alleys: **100-150 mm**
 - Grade: **1.25- 5%**

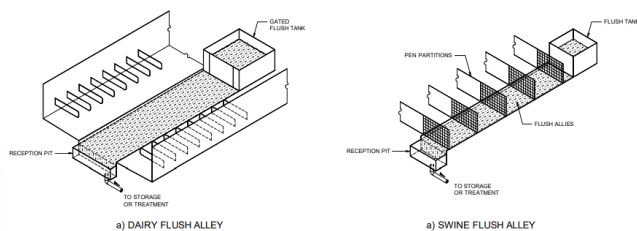


Figure 4 – Typical flush alley

Gutters

- **Gravity drain gutter**
 - Depth: **760 mm**

- Width: **150 mm**
- Slope: **1% towards the drain**

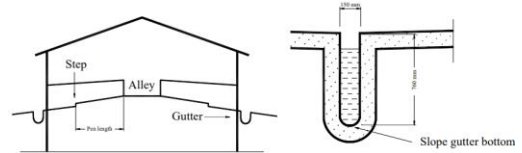


Figure 5 – Gravity drain gutter for swine manure

• Step-dam gutters

- Collecting cattle manure
- High: **150 mm**
- Wide: **760 mm**

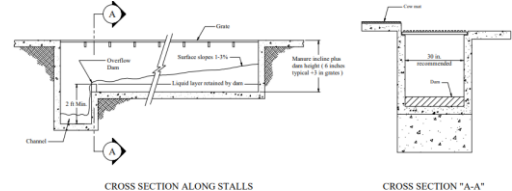


Figure 6 – Step-dam gutter for dairy manure

• Scrape gutters

- Confined stall barns
- Wide: **0.4- 0.6 m**
- Deep: **0.3- 0.4 m**

• Flush gutters

- Depth: **0.6 m**
- Bottom slope: **0-5%**

RUNOFF COLLECTION

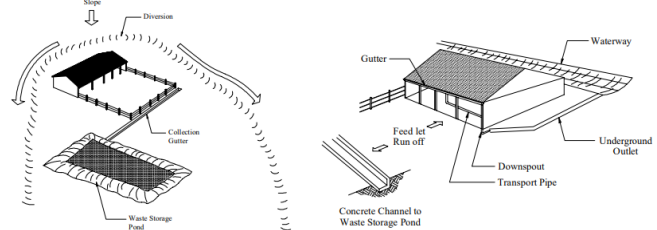


Figure 7 – Typical layout of collecting waste

- Bottom slope: **0.5%**

RECEPTION PITS

- designed to hold all the waste for several days.

SCREENING

• Coarse screening

Table 1 – Description of coarse screening devices used in liquid waste treatment

CLASSIFICATION OF SCREENING DEVICE	SIZE CLASSIFICATION	SCREEN OPENING mm	SCREEN MATERIAL	APPLICATION
Bar screen	Coarse	25 – 50	Bars	Removal of coarse suspended solids and as pretreatment step for fine screening.
Manually cleaned	Coarse	15 – 75	Bars	
Mechanically cleaned				
Fine bar or perforated coarse screen (mechanically cleaned)	Fine coarse	3 – 12.5	Thin bars	Pretreatment
Fine bar	Fine coarse	3 – 9.5	Perforated plate	Pretreatment
Perforated plate	Fine coarse	3 – 12.5	Stainless steel wedge wire	Pretreatment
Rotary drum				

PAES: AGRICULTURAL STRUCTURES

Table 2 – Typical design specification for manually and mechanically cleaned bar racks

ITEM	UNIT	MANUALLY CLEANED	MECHANICALLY CLEANED
Bar size:			
Width	mm	5 – 15	5 – 15
Depth	mm	25 – 38	25 – 38
Clear spacing between bars	mm	25 – 50	15 – 75
Slope from vertical	%	58 – 100	0 – 58
Approach velocity	m/s	0.3 – 0.6	0.6 – 1
Allowable headloss	mm	152	152

• Fine screening

Table 3 – Description of fine screening devices used in liquid waste treatment

CLASSIFICATION OF SCREENING DEVICE	SIZE CLASSIFICATION	SCREEN OPENING mm	SCREEN MATERIAL	APPLICATION
Fine screen (mechanically cleaned)				
Fixed parabolic	Fine	0.25 – 3.2	Stainless steel wedge wire	Pretreatment
Rotary drum	Fine	0.25 – 3.2	Stainless steel wedge wire	Pretreatment
Rotary disk	Very fine	0.15 – 0.38	Stainless steel wedge wire	Primary treatment

SIZE REDUCTION

- should be provided to cut solids in the liquid waste passing through the device to about **6 mm**
- Head loss through the comminutor should range from **0.3 m – 0.9 m**

GRIT CHAMBERS

- should be designed to remove particles equivalent to fine sand (**0.25 mm – 3.2 mm diameter with a specific gravity of 2.65**).
- Horizontal-flow grit chamber**

Table 4 – Typical design information for horizontal-flow grit chambers

ITEM	UNIT	VALUE
Detention time	s	60
Horizontal velocity	m/s	0.3
Settling velocity for removal of:		
50-mesh material ^a	m/s	0.05
100-mesh material ^a	m/s	0.01
Headloss in a control section as percent of depth in channel	%	36 ^b
Added length allowance for inlet and outlet turbulence	%	30
^a If the specific gravity of the grit is significantly less than 2.65, lower velocities should be used.		
^b For Parshall flume control.		

• Aerated grit chambers

Table 5 – Typical design information for horizontal-flow grit chambers

ITEM	UNIT	VALUE
Detention time at peak flowrate	min	3
Dimensions:		
Depth	m	3
Length	m	12
Width	m	3.5
Width-depth ratio		1.5:1
Length-width ratio		4:1
Air supply per meter of length	m ³ /m-min	0.46
Grit quantities	m ³ /Mgal	0.06

• Vortex-type grit chambers

Table 6 – Typical design information for vortex-type grit chambers

ITEM	UNIT	VALUE
Detention time at peak flow rate	s	30
Diameter:		
Upper chamber	m	1.2 – 7.3
Lower chamber	m	0.9 – 1.8
Height	m	2.7 – 4.9
Removal rates:		
50 mesh material	%	92 – 98
70 mesh material	%	80 – 90
100 mesh material	%	60 – 70

SEDIMENTATION TANKS

- Either rectangular or circular
- Rectangular width ratio: **3:1- 5:1**

Table 7 – Typical design information for sedimentation tanks for primary and secondary treatment of liquid waste

ITEM	UNIT	VALUE	
		PRIMARY	SECONDARY
Rectangular:			
Depth	m	3 – 4.9	3 – 6.7
Length	m	15 – 90	15 – 90
Width	m	3 – 24	3 – 24
Flight speed	m/s	0.01 – 0.2	0.01 – 0.02
Circular:			
Depth	m	3 – 4.9	3 – 6.7
Diameter	m	3 – 60	3 – 60
Bottom slope	%	6 – 17	6 – 17
Flight travel speed	rev/min	0.02 – 0.05	0.02 – 0.05

Table 8 – Typical detention time and overflow rates for primary sedimentation tanks

ITEM	UNIT	VALUE
Primary sedimentation tanks followed by secondary treatment		
Detention time	h	1.5 – 2.5
Overflow rate		
Average flow	m ³ /m ² -d	30 – 50
Peak hourly flow	m ³ /m ² -d	81 – 122
Weir loading	m ³ /m-d	124 – 497
Primary settling with waste activated sludge return		
Detention time	h	1.5 – 2.5
Overflow rate		
Average flow	m ³ /m ² -d	24 – 33
Peak hourly flow	m ³ /m ² -d	49 – 69
Weir loading	m ³ /m-d	124 – 497

SETTLING CHANNEL

- Side slope: **3:1**
- Bottom slope: **0.1- 0.3%**
- For the first **15 m – 30 m** section, **0.6 m/s** velocity should be designed to settle out relatively large solids and debris. For the next **30 m – 91 m**, the velocity should be as low as **0.15 m/s** to settle out smaller solids.

OIL AND GREASE INTERCEPTOR

- Volume should be **1-3 times** the average daily flow rate

STORAGE GUTTERS

- Depth: **760 mm**
- Width: **150 mm**
- Bottom slope: **1%**

STORAGE TANKS

- Covers for tank opening should be non-floating and weigh at least **18 kg**

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ANAEROBIC LAGOONS

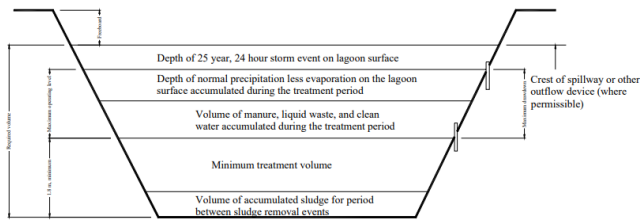


Figure 8 – Anaerobic lagoon cross-section

Waste Volume

$$WV = VM_T + VWW_T + CW$$

Minimum Treatment Volume

$$TV_{\min} = \frac{VS_T}{VSLR}$$

Sludge Volume

$$SV = 365 \times AU \times TS \times SAR \times T$$

Table 9 – Sludge accumulation ratio (SAR)

Animals	SAR
Poultry	0.0295
Swine	0.0485
Cattle	0.0729

- Freeboard: **0.3 m**
- Minimum depth: **1.8 m**

AEROBIC LAGOONS

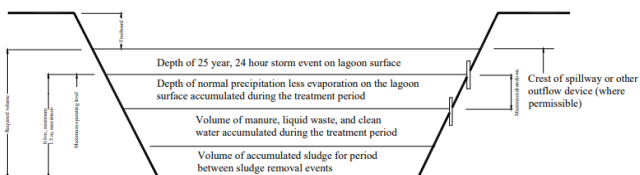


Figure 9 – Aerobic lagoon cross-section

Waste Volume

$$WV = VM_T + VWW_T + CW$$

Sludge Volume

$$SV = 365 \times AU \times TS \times SAR \times T$$

Table 9 – Sludge accumulation ratio (SAR)

Animals	SAR
Poultry	0.0295
Swine	0.0485
Cattle	0.0729

Minimum Treatment Surface Area

$$TSA_{\min} = \frac{BOD_T}{BOD_{LR}}$$

- Min depth: **0.6 m** ; max depth: **1.5 m**
- Freeboard: **0.3 m**

EARTH EMBANKMENT

- Top width: **2 m**
- Inside slopes: **1:2 and 1:4**
- Outside slope: **1:5**
- Elevation increase: **5%**

- The top of the settled embankment should be at least **0.6 m** above the maximum design surface level.

INLET AND OUTLET

- Pipes or open channels: **200 mm** dimension
- Outlet devices: allows effluent to be taken at a level **150 mm – 450 mm** below the surface.

WATER SUPPLY

- Maintain operating level of the lagoon at **60%** of the design depth

FREE-WATER-SURFACE CONSTRUCTED WETLANDS

- emergent vegetation should be flooded to a depth of **100 mm – 450 mm**
- basin slope: **0.4-0.5%**
- Detention time: **2-5 days**

SUBSURFACE-FLOW CONSTRUCTED WETLANDS

- The gravel size for the bed should range from **3 mm – 32 mm**. For the inlet zone, gravel size should be **50 mm – 100 mm** to minimize clogging.
- depth of the bed should be **0.45 m – 1 m** with a slope of **0 % - 0.5 %**
- water level should be kept **75 mm – 150 mm** below the top of the medium.
- Detention time: **3-4 days**

FLOATING AQUATIC PLANT SYSTEM

Table 10 – Typical design criteria for water hyacinth wastewater treatment system

ITEM	UNITS	TYPE	
		AEROBIC NONAERATED	AEROBIC AERATED
Influent wastewater		Fine-screened or settled	Fine-screened or settled
Water depth	m	0.45 – 0.75	1.2 - 1.4
Detention time	days	10 -30	4 – 8
Aeration	m ³ /min.Mgal	none	11.32– 12.03
Type of aerator		none	Fine bubble
Channel cross section		Trapezoidal	Trapezoidal
Channel top width	m	6 - 9	6 - 9
Channel side slopes		1:1	1:1
Channel lining		Geomembrane	Geomembrane
Liner thickness	mil	40 – 80	40 – 80
Pond geometry		Horseshoe shape	Horseshoe shape

Table 11 – Typical design criteria for duckweed system

ITEM	UNIT	VALUE
Influent wastewater		Facultative pond influent
Detention time	days	20 – 30
Water depth	m	1.5 – 2.4

EFFLUENT TREATMENT

Filtration

- width-to-length ratios for filter bed should **1:1 – 1:4**

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- Filter medium such as sand should have an effective diameter of at least **0.45 mm** and shall have a minimum depth of **280 mm**.

SLUDGE TREATMENT

Drying

- The bed should consist of **300 mm** fine sand underlaid by **200 mm – 460 mm** of gravel.
- The sand should have an effective size of **0.3 mm – 0.75 mm**, free from fines and should have a maximum uniformity coefficient of **3.5**. The gravel size should be **2.5 mm – 25 mm**.
- Underdrains should be **100 mm** in diameter and should slope at **1%**.
- Beds should be divided every **3 m** by **0.6 m** partitions.
- Sludge drying lagoon should have a sludge depth of **0.62 m – 1.25 m**. Storage lagoon should have a depth of **3 m – 5 m** and should be provided with mechanical aerator.

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Agricultural Structures – Waste Management Structures Part 2: Agricultural Solid Waste

agricultural solid waste

wastes resulting from the production and processing of crops and animals or agricultural products, including manures with at least 20% solids, pruning and crop residues wherever produced

bulking agent

any item used to improve the compost structure and to increase porosity to allow internal air movement

C:N ratio

weight ratio of carbon to nitrogen

compost mix

mixture of an organic waste with amendment(s) or bulking agent(s) in the proper proportions to promote aerobic microbial activity and growth and to achieve optimum temperatures

composting

controlled decomposition of organic matter by micro-organisms, mainly bacteria and fungi, into a humus-like product

composting amendment

any item added to the compost mixture that alters the moisture content, C:N ratio, or pH

curing

process where fungi digest the carbons not degraded during composting and further stabilize the nutrients

disposal site

site where solid waste is finally discharged and deposited

hazardous waste

solid waste or combination of solid waste which because of its quantity, concentration, or physical, chemical or infectious characteristics may: cause, or significantly contribute to an increase in mortality or an increase in serious irreversible, or incapacitating reversible, illness

leachate

shall refer to the liquid produced when waste undergo decomposition, and when water percolate through solid waste undergoing decomposition; contaminated liquid that contains dissolved and suspended materials

storage

interim containment of solid waste after generation and prior to collection for ultimate recovery or disposal

windrow composting

involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions

FLOOR

- Slope: **2%**
- Floor drains: **150 mm diameter**
- Ramp slope: **20:1- 10:1**

COMPOSTING TIME

Table 1 – Typical composting time

Method	Composting time
Windrow	4 months
Aerated static pile	4 weeks
In-vessel	7 – 30 days

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C:N RATIO

- Compost mix: between **25:1 and 40:1**

Table 2 - Typical carbon to nitrogen ratios of common composting materials

Material	C:N ratios
Cattle manure (with straw)	25-30
Cattle feedlot	13
Corn & sorghum stover	60-100
Cucumber	20
Dairy manure	10-18
Garden wastes	20-60
Grain rice	36
Grass clippings	12-25
Green leaves	30-60
Horse	19
Lamb	10
Pig manure	5-8
Potato tops	25
Poultry manure (fresh)	6-10

Material	C:N ratios
Poultry manure (henhouse litter)	12-18
Residue of mushroom culture	40
Rice straw	48-115
Saw dust	300-723
Seaweed	19
Soybean residues	20-40
Straw	40-80
Sugar cane (trash)	50
Tomato leaves	13
Tomatoes	25-30
Veal	2
Watermelon	20
Water hyacinth	20-30
Weeds	19
Wood chips	100-441

pH

- Compost mixture: **5.5- 8.0**

AEROBIC CONDITIONS

- 5% oxygen**

TEMPERATURE CONTROL

- Fully enclosed vessel: 55C** on at least **3 different days**
- Windrows/ not fully enclosed: 55C** on at least **15 different days**
- Static pile method: 55C** on at least **15 different days**

MOISTURE

- 50-60%**

PARTICLE SIZE

- 5-50 mm**

WINDROW COMPOSTING

- 1 m – 2 m high** and **2 m – 5 m wide** at the base

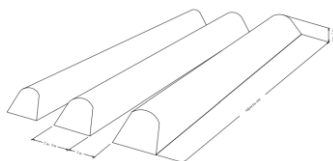


Figure 1 – Typical windrow composting

AERATED STATIC PILE

- The compost mixture height generally ranges from **2.4 m – 4.5 m** and the width should be twice the depth.
- Individual piles should be spaced about a **half** the distance of the height.

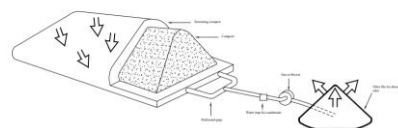


Figure 2 – Typical static pile composting

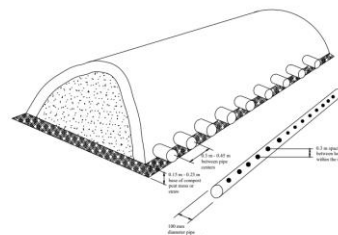


Figure 3 – Typical duct placement for static pile composting

CURING

- 1-2 months**

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Agricultural Structures – Greenhouse

greenhouse

structure that provide a reliable enclosure within which an environment favorable to plant growth can be attained

ridge and furrow greenhouse

gutter connected method of greenhouse construction where modular roof units are connected at the eave by a common gutter

cooling

removal of heat from the interior of the greenhouse

heating

addition of heat to the interior of the greenhouse from any energy source including the sun

shading

prevents excess solar radiation in the greenhouse

ventilation

process of exchanging air inside the greenhouse with outside air to control temperature, humidity, oxygen or carbon dioxide levels

mechanical ventilation

ventilation of greenhouse using fans

natural ventilation

ventilation of greenhouse which occurs through controlled openings due to natural pressure variations inside and outside the greenhouse

PAES: AGRICULTURAL STRUCTURES

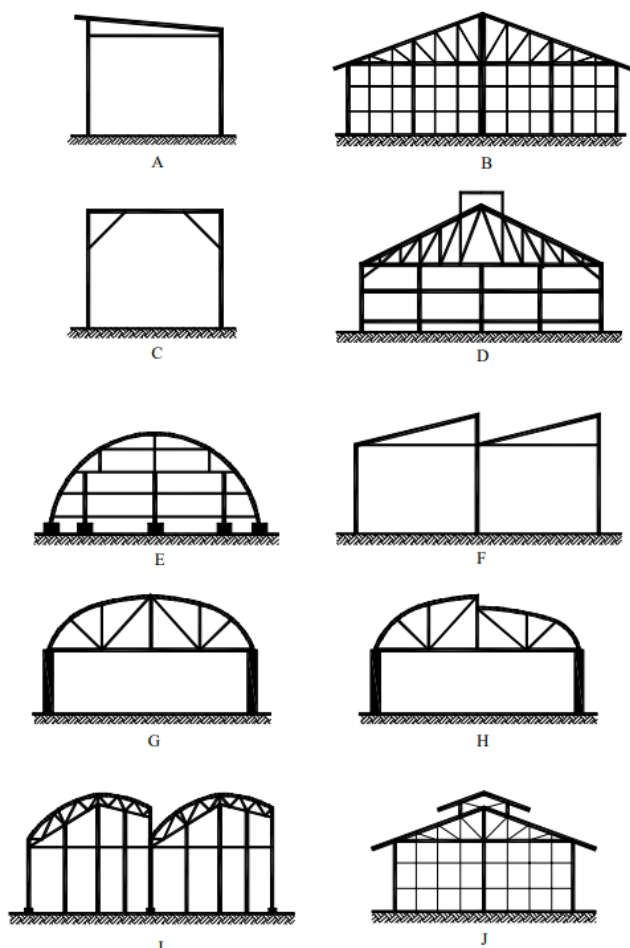


Figure 1. Roof shape: a) Lean-to; b) Gable; c) Flat; d) Monitor; e) Quonset; f) Saw tooth; g) arch; h) Uneven arch; i) Arch saw and j) Gable

LOCATION

- **North to South** Orientation

BENCH AREA REQUIREMENT

- should be up to **1.8 m wide** if they are to be access from both sides, and should not exceed **0.90 m** if they are accessible only from one side.
- Air circulation space: **160 mm**
- Usable space
 - Fixed bench: **60-75%**
 - Rolling benches: **90%**

PLANT REQUIREMENT

Table 2 – Population density per square meter of bench based on pot sizes

Pot size ^a	Population density per m ² of bench ^b	Pot size ^a	Population density per m ² of bench ^b
10	100	170	36
11	81	180	25
12	64	200	25
13	42	230	16
15	36	250	16
16	36	300	16

^a pot size based on diameter of pots commercially available, in mm
^b population in dense packing

Table 3 – Population density at recommended plant spacing for vegetables

Vegetable	Recommended plant spacing	Population density per m ² of bench
Sweet pepper	30 cm between hills (2 rows bed)	6-8
Chili	35 cm between hills (2 rows bed)	6-8
Tomato	40 cm between hills (2 rows bed)	4-6
Sweetmelon	50-60 cm between hills (2 rows bed)	4
Watermelon	50-60 cm between hills (2 rows bed)	4
Cucumber	50-60 cm between hills (2 rows bed)	4
Lettuce	15 cm between hills (4 rows bed)	24-28
Broccoli	30 cm between hills (3 rows bed)	9-12

Table 4 – Population density at recommended plant spacing for ornamentals

Ornamentals	Recommended plant spacing		Population density per m ² of bench
	Between hills mm	Between rows mm	
Baby Aster	100 - 150	250	28-40
Chrysanthemum	120 - 250	250	16-32
Daisy	250	360 - 460	8-12
Gladiolus	60 - 80	200	60-85
Rose	300	300	9
Orchids (Vanda)	10 plants/driftwood		1-2 driftwood

WALKWAYS

- Determinate crops: **0.30 m**
- Indeterminate crops: **0.80 m**
- Secondary aisle: **0.60-0.90 m**
- Primary aisle: **1.2-1.8 m**
- If in wheelchair: **1.2 m** aisle with ramps

FLOOR LENGTH AND WIDTH

- Length shall be determined by the number of plants to be grown
- For plastic structure, extra **0.60 m** to fasten the plastic properly
- Length of greenhouse: **50 m**
- Gutter connected greenhouse: **50 m total width**

BUILDING HEIGHT

- Eave height: **1.70 m**
- Gable height: **2.4 m**
- The height of the structure shall be equal to the eave height plus **one-fourth** the width.

WORKING HEIGHT

- Gutter height: **2.8- 3.0 m**
- Height of the house in the walk areas: **2.0 m**
- For tall crops **2.0 m** shall be the minimum height at the eaves and **2.10 m** as minimum desirable height.

FOUNDATION

- Permanent greenhouse: shall extend to a minimum **depth** of **0.45 m**

WALLS

- Height of side posts and columns should be **0.30 m - 3 m**.
- greenhouses with wooden structural members, a **0.4 m** high reinforced

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concrete wall at the bottom shall be provided for support.

ROOF

Frames

- able to carry trellising loads up to **25 kg/m²**
- able to withstand wind gusts of **250 km/hr**

Framing materials

- **Wood**
 - Rafters: **0.6- 1.2 m** from center to center
 - Purlins: **2.5 m** spacing

Covering materials

- **Glass**
- **Plastic**
 - Polyethylene
 - Fiberglass
 - Corrugated plastic sheets
 - Screen

Roof Pitch

- Glass greenhouse: **51%**
- Plastic covered greenhouses: **58-70%**

Gutter

- Slope: **2%**

ACCESS

- If walking tractors are to be used, **1.2 m** wide standard doors shall be provided. A minimum of **0.8 m** wide side doors shall be used.

FLOOR AND WALKWAYS

- A concrete, gravel or stone walkway **0.60 m – 0.9 m** wide should be built for easy access to the plants.

BENCHES

- Bench size: **0.90 m high**
- For temporary benches for growing and bedding plants, benches as wide as **1.50 m - 1.80 m** and only **0.45 m - 0.60 m** high should be used.
- When shade plants are to grow under the benches, bench heights as high as **1.20 m** should be used.

ARTIFICIAL LIGHTING

- **200 lux**
- Use fluorescent instead of incandescent

IRRIGATION

- Water should be applied by means of a overhead sprinkler, a drip or trickle irrigation system, by hand using a hose or

similar device, or some combination of these delivery system.

- Drip or trickle systems are the most efficient and provide greater control over the amount of water applied.
- general rule of thumb to follow in irrigating greenhouse crops is to apply 10%- 15% more water than the container will hold to facilitate leaching at each irrigation and reduce the potential for the accumulation of soluble salts.
- Drainage rate; 30- 50%
- Soil beds and benches slope: 0.8-1.0 %

VENTILATING AND COOLING GREENHOUSE

Natural ventilating system

- Large vent openings provide the most ventilation, the total vent area should be **15%- 25%** of the floor area.
- Top hinged ridge vents: **60 degrees angle** with roof when fully opened

MECHANICAL VENTILATING SYSTEM

- Ventilation fans: **0.03 kPa** static pressure
- Fan spacing: not more than **7.5 m** apart
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